COMP4131: Data Modelling and Analysis

Lecture 2: Data Wrangling and Pre-processing

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March 4, 2026

Outline

1. Introduction to Data Wrangling and Pre-processing
2. The Data
3. Data Cleaning
4. Data Transformation
5. Data Integration and Reduction
6. The Complete ML Workflow

Introduction to Data Wrangling and Pre-processing

Data Wrangling and Pre-processing 数据整理和预处理

Data wrangling and data pre-processing are interconnected processes in the data preparation pipeline that involve cleaning, transforming, and organizing raw data into a structured and optimized format suitable for analysis, modeling, or decision-making.
数据整理和数据预处理是数据准备流程中相互关联的两个过程，涉及对原始数据进行清洗、转换和组织，使其成为适合分析、建模或决策的结构化和优化格式。

Data Wrangling and Pre-processing

- Data wrangling is the process of cleaning, transforming, and organizing raw data into a usable format.
- It involves handling inconsistencies, missing values, and errors, as well as integrating data from multiple sources to prepare it for analysis or further processing.

Workflow of the Data Wrangling Process

Data Wrangling and Pre-processing

- Data pre-processing is the process of preparing data for analysis or machine learning after it has been wrangled.
- It ensures the data is optimized for specific analytical or modeling purposes via techniques such as:
  - Scaling / normalization 缩放/归一化
  - Encoding categorical variables
  - Handling outliers 处理离群值
  - Splitting data into training and testing sets

Why is Data Wrangling and Pre-processing Important?

Real-World Data Challenges:

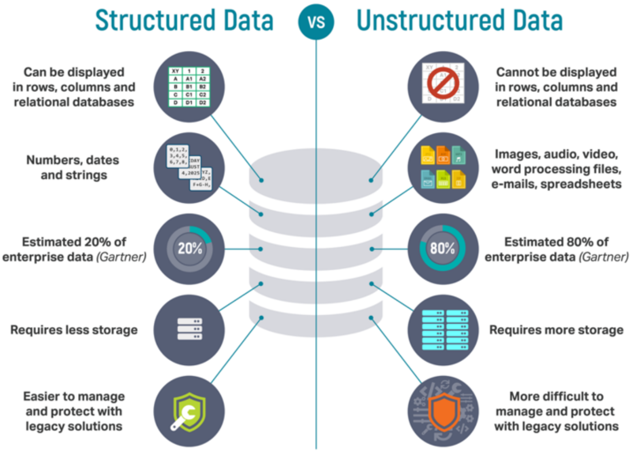
- **Missing values:** Incomplete data that can lead to biased results.
- **Duplicates:** Repeated records that inflate or skew analysis.
  夸大或扭曲分析结果
- **Inconsistent formats:** Non-standardized data, such as mixed date formats or varying units.单位不统一
- **Outliers:** Extreme values that can distort statistical calculations.
  异常值：极端值会扭曲统计计算。

Impact on Analysis:

- Poor-quality data leads to **inaccurate insights** and **unreliable models**.低质量数据会导致不准确的洞察和不可靠的模型
- **Garbage in, garbage out (GIGO):** Results are only as good as the data used. 垃圾进，垃圾出 (GIGO)：结果的质量取决于所使用的数据质量

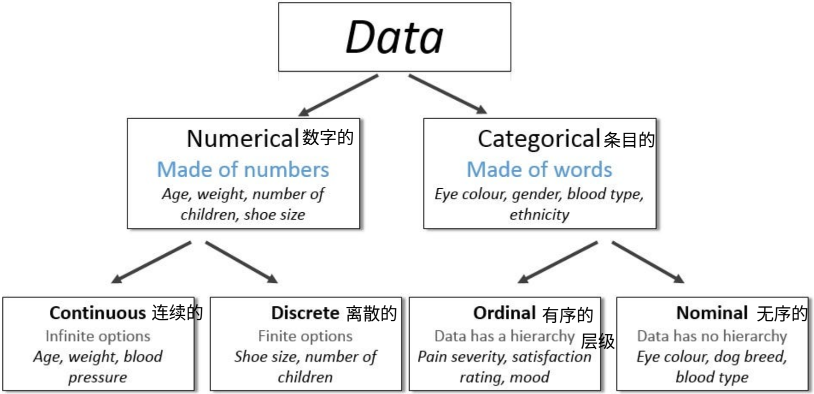
The Data

Types of Data



Source: Lawtomated, Structured vs. Unstructured Data: What are they and why care?

Types of Data



Source: Medium: Data Science Basics.

Common Notations 常用符号

Feature (Attribute)

Label (Class, Output, Target)

Examples (Samples, Instances, Observations)

housingMedianAge (feature)	totalRooms (feature)	totalBedrooms (feature)	medianHouseValue (label)
15	5612	1283	66900
19	7650	1901	80100
17	720	174	85700
14	1501	337	73400
20	1454	326	65500

Feature and Label

Features:

- A feature is an input variable—the x variable in machine learning.
- A simple machine learning project might use a single feature, while a more sophisticated machine learning project could use millions of features, specified as:

$$x_1, x_2, \ldots, x_N$$

Labels:

- A label is the thing we're predicting—the y variable in machine learning.

Examples

Examples:

- An **example** is a particular instance of data, $\mathbf{x}$. (We put $\mathbf{x}$ in boldface to indicate that it is a vector.)
- Examples can be categorized into:
  - **Labeled examples** - Includes both feature(s) and the label. That is:

    Labeled examples: {features, label} : $(\mathbf{x}, y)$

  - **Unlabeled examples** - Contains features but not the label. That is:

    Unlabeled examples: {features} : $(\mathbf{x})$

Data Cleaning

Handling Missing Values

What are Missing Values?

- Missing data can occur due to errors in data collection, storage, or processing.

Impact:

- Missing data can lead to biase or incorrect analysis.

	A	B	C	D
0	1.0	2.0	3.0	4.0
1	5.0	6.0	NaN	8.0
2	10.0	11.0	12.0	NaN

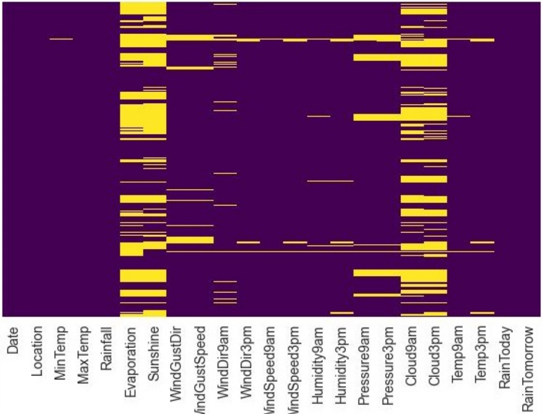
Identifying Missing Data

Methods to Identify Missing Data:

- Visual inspection (e.g., heatmaps).可视化：热力图
- Programmatic methods: `isnull()` or `isna()` in Pandas.程序化

Example:

- A dataset with missing values in columns.
  热力图中“亮色块”表示缺失分布，可快速看缺失是否集中在某些列/某些时间段



Handling Missing Values

Common Techniques:

- Remove missing values (rows/columns).
- Impute missing values (mean, median, or mode).
  插补
- Forward/backward fill: use previous or next values to fill missing data.
  前向/后向填充

	A	B	C	D
0	1.0	2.0	3.0	4.0
1	5.0	6.0	NaN	8.0
2	10.0	11.0	12.0	NaN

Remove rows

	A	B	C	D
0	1.0	2.0	3.0	4.0

Remove cols

	A	B
0	1.0	2.0
1	5.0	6.0
2	10.0	11.0

Handling Missing Values

Mean Imputation: 均值插补

$$x_{\text{new}} = \frac{\sum x_i}{n}$$ 计算该列的平均值

	A	B	C	D
0	1.0	2.0	3.0	4.0
1	5.0	6.0	NaN	8.0
2	10.0	11.0	12.0	NaN

Imputation

	A	B	C	D
0	1.0	2.0	3.0	4.0
1	5.0	6.0	7.5	8.0
2	10.0	11.0	12.0	6.0

Outlier Detection and Treatment

What are Outliers?

- A data point that significantly deviates from other observations.
  与其他观测值存在显著偏差的数据点。

Impact:

- Outliers can distort statistical analysis and model performance.

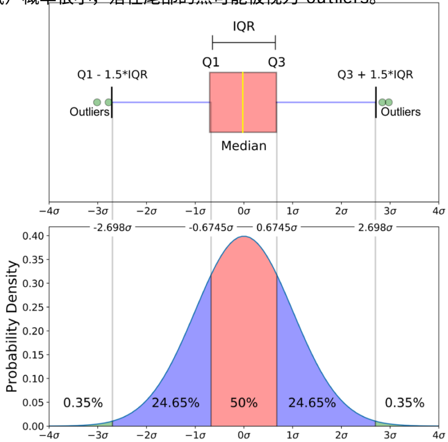
Methods to Identify Outliers:

- Z-score.
- IQR (Interquartile Range).

Z-score:

- Measures how far a data point is from the mean in terms of standard deviations. 标准差
- Formula: $Z = \frac{(x-\mu)}{\sigma}$

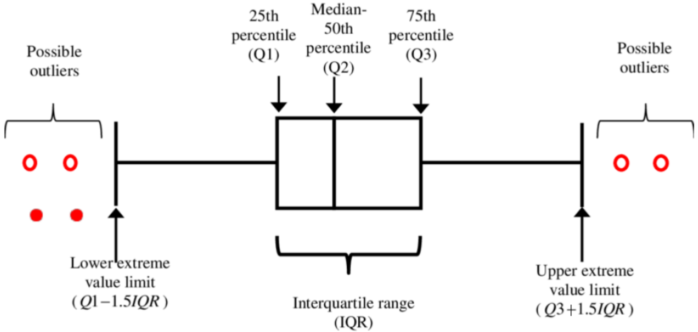
正态分布尾部（极端区域）概率很小，落在尾部的点可能被视为 outliers。



Methods to Identify Outliers

Interquartile Range (IQR):

- Identifies outliers as data points outside 1.5 $\times$ IQR.
- Formula: $\mathrm{IQR} = Q3 - Q1$



Handling Outliers

删不删取决于“它是错误还是稀有但真实的现象”。

Remove:

- Delete outlier rows.

Cap:

- Replace outliers with a threshold value (e.g., upper/lower bounds).

Transform:

- Apply log or square root transformations to reduce the impact of outliers.

Dealing with Duplicates

What are Duplicates?

- Duplicates can occur due to data entry errors or merging datasets.

Impact:

- Duplicates can lead to overcounting and biased results.

Methods:

- Identify and remove duplicates.

Data Transformation

Introduction to Data Transformation

What is Data Transformation?

- The process of converting data into a suitable format for analysis or modeling.

Why is it Important?

- Improves the performance of the models.
- Ensures data is on a consistent scale.
- Handles categorical and continuous data appropriately.

Key Techniques:

- Feature scaling.
- Encoding categorical data.
- Data binning. 数据分箱。
- Feature engineering.
- Handling imbalanced data.

Feature Scaling

What is Feature Scaling?

- Rescaling features to a specific range (e.g., 0 to 1) or standardizing them to have a mean of 0 and a standard deviation of 1.
  将特征重新缩放到特定范围（例如，0 到 1）或将其标准化，使其均值为 0，标准差为 1。

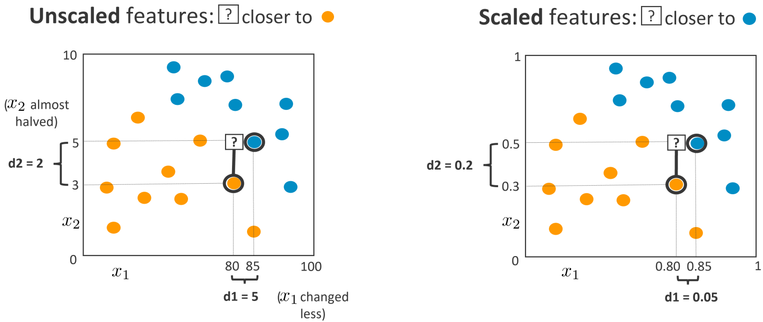
Why is it Important?

- Ensures all features contribute equally to the model.
- Prevents features with larger scales from dominating.
  防止规模较大的特征占据主导地位。

Common Methods:

- Normalization (Min-Max scaling).
- Standardization (Z-score scaling).

Feature Scaling



Source: AWS Machine Learning University (MLU)

Normalization vs. Standardization

Normalization (Min-Max Scaling):

- Rescales data to a range of [0, 1].
- Formula: $X_{\text{norm}} = \frac{X - X_{\text{min}}}{X_{\text{max}} - X_{\text{min}}}$

Standardization (Z-score Scaling):

- Rescales data to have a mean of 0 and a standard deviation of 1.
- Formula: $X_{\text{std}} = \frac{X - \mu}{\sigma}$

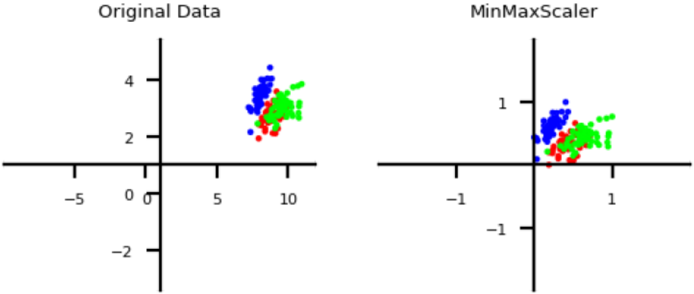
When to Use:

- Normalization: When data distribution is unknown or not Gaussian.
  当数据分布未知或不服从高斯分布时。
- Standardization: When data follows a Gaussian distribution.
  当数据服从高斯分布时。

Normalization

Normalization (Min-Max Scaling):

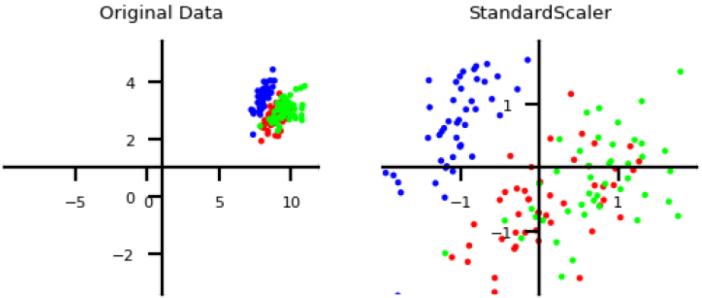
- Scales all features between a given min and max value (e.g. 0 and 1).
- Makes sense if min/max values have meaning in your data.
- Sensitive to outliers.



Standardization

Standardization (Z-score Scaling):

- Generally most useful, assumes data is more or less normally distributed.
- Per feature, subtract the mean value μ, scale by standard deviation σ



Encoding Categorical Data

What is Categorical Data? made of word

- Data that represents categories (e.g., gender, color, country).

Why Encode Categorical Data?

- Most machine learning algorithms require numerical input.

Common Encoding Techniques:

- One-hot encoding.
- Label encoding.
- Binary encoding.

One-hot Encoding, Label Encoding, Binary Encoding

label encoding 可能引入“虚假大小关系”（对无序 nominal 类别危险）
one-hot 维度高但语义更安全
binary encoding 折中（维度更低）

One-hot Encoding:

- Converts each category into a binary vector.
- Example: Red = [1, 0, 0], Green = [0, 1, 0], Blue = [0, 0, 1].

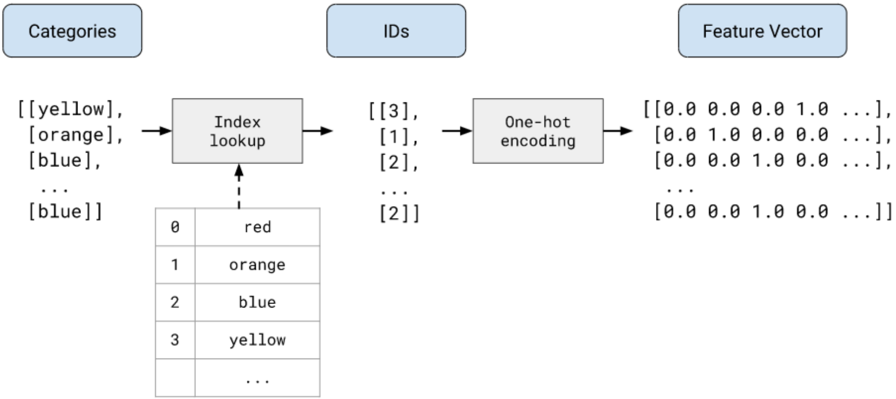
Label Encoding:

- Assigns a unique integer to each category.
- Example: Red = 0, Green = 1, Blue = 2.

Binary Encoding:

- Combines label encoding with binary representation.
- Example: Red = 00, Green = 01, Blue = 10.

One-hot Encoding



Data Binning

What is Data Binning?

- Converting continuous data into discrete categories (bins).
  把连续值离散化成区间类别

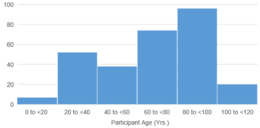
Why Use Data Binning?

- Simplifies complex data.
- Handles outliers.
- Improves model performance for certain algorithms.

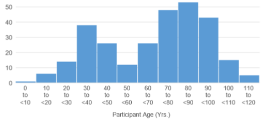
Example (Age):

- 0-18 = "Child",
- 19-35 = "Young Adult",
- 36-60 = "Adult",
- 60+ = "Senior".

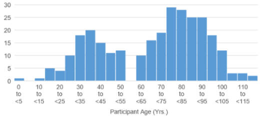
Data Binning



6 bins



12 bins



24 bins

Feature Engineering

What is Feature Engineering?

- The process of creating new features or modifying existing ones to improve model performance.

Techniques:

- Adding new features (e.g., calculating ratios or differences).
- Modifying features (e.g., log transformation).
- Dropping irrelevant or redundant features. 删除不相关/冗杂特征

Example:

- Creating a "BMI" feature from height and weight.

What is Imbalanced Data?

- When one class significantly outnumbers the other(s) in a classification problem. 当一个类别的数量明显多于其他类别时，分类问题就会出现这种情况。

Why is it a Problem?

- Models may become biased toward the majority class.

Techniques to Handle Imbalanced Data:

- Resampling: Oversampling the minority class or undersampling the majority class. 重采样：对少数类进行过采样或对多数类进行欠采样。
- Synthetic Data Generation: Using techniques like SMOTE, GAN.
  合成数据生成
- Algorithmic Approaches: Using class-weighted models.
  算法方法：使用类别加权模型

ACTIVITY . THINK-PAIR-SHARE

- Pair up with the person next to you and identify what are the issues in the following heart disease classification data set. What are the solutions?

PatientID	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS	RestingECG	MaxHR	ExerciseAngina	Oldpeak	ST_Slope	HeartDisease
1	52	M	TA	118	186	0	LVH	190	N	0	Flat	0
2	58	M	ASY	136	203	1	Normal	123	Y	1.2	Flat	1
3	44	M	ASY	110	197	0	LVH	177	N	0	Up	1
4	43	M	TA	120	291	0	ST	155	N	0	Flat	1
5	55	F	ATA	132	342	0	Normal	166	N	1.2	Up	0
6	66	M	ASY	112	212	0	LVH	132	Y	0.1	Up	1
7	55	M	NAP			0	Normal	155	N	1.5	Flat	1
8	53	M	ASY	123	282	0	Normal	95	Y	2	Flat	1
9	42	M	ASY	136	315	0	Normal	125	Y	1.8	Flat	1
10	43	M	ASY	110	211	0	Normal	161	N	0	Up	0
11	59	M	ASY	125		1	Normal	119	Y	0.9	Up	1
12	46	M	ASY	140	311	0	Normal	120	Y	1.8	Flat	1
13	42	M	ATA	120	198	0	Normal	155	N	0	Up	0
14	39	M	ASY	110	273	0	Normal	132	N	0	Up	0
15	50	F	ASY	110	254	0	LVH	159	N	0	Up	0
16	60	M	ASY	142	216	0	Normal	110	Y	2.5	Flat	1
17	56	M	ASY	115		1	ST	82	N	-1	Up	1
18	44	F	NAP	118	242	0	Normal	149	N	0.3	Flat	0
19	60	M	ASY	136	195	0	Normal	126	N	0.3	Up	0
20	32	M	ATA	125	254	0	Normal	155	N	0	Up	0
21	58	M	ATA	125	220	0	Normal	144	N	0.4	Flat	0
22	37	F	NAP	120	215	0	Normal	170	N	0	Up	0
23	57	M	ASY	140		1	Normal	100	Y	0	Flat	1
24	69	M	ASY	145	289	1	ST	110	Y	1.8	Flat	1
25	53	F	NAP	128	216	0	LVH	115	N	0	Up	0
26	44	M	ATA	120	184	0	Normal	142	N	1	Flat	0
27	34	M	ATA	98	220	0	Normal	150	N	0	Up	0
28	77	M	ASY	125	304	0	LVH	162	Y	0	Up	1
29	74	F	ATA	120	269	0	LVH	121	Y	0.2	Up	0
30	47	M	NAP	108	243	0	Normal	152	N	0	Up	1
31	63	M	ASY	96	305	0	ST	121	Y	1	Up	1
32	56	M	NAP	155		0	ST	99	N	0	Flat	1
33	32	M	TA	95		1	Normal	127	N	0.7	Up	1
34	65	F	ASY	150	225	0	LVH	114	N	1	Flat	1
35	65	M	ASY	144	312	0	LVH	113	Y	1.7	Flat	1

Data Integration and Reduction

Data Integration and Reduction

What is Data Integration?

统一

- The process of combining data from different sources into a unified dataset.
- Ensures consistency and usability for analysis.

What is Data Reduction?

- The process of reducing the size or complexity of the dataset.
- Aims to retain important information while improving efficiency.

保留重要信息的同时提高效率

Key Techniques:

- Combining datasets (merging, concatenation, joining).
- Feature selection.
- Dimensionality reduction (PCA, t-SNE, UMAP).

Combining Datasets

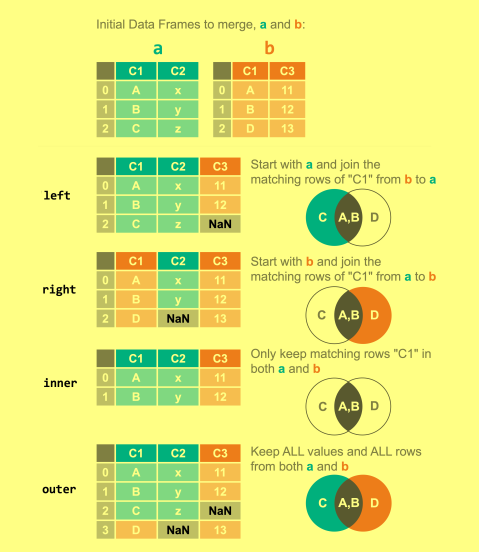
Why Combine Datasets?

- To enrich data by adding more features or samples.

Common Techniques:

- Merging: Combines datasets based on common keys (e.g., primary keys in databases).
- Concatenation: Stacks datasets either row-wise or column-wise.
  按行/列堆叠
- Joining: Combines datasets similar to SQL joins (e.g., inner, outer, left, right joins).

Combining Datasets



left join：保留左表全部行，右表能匹配的补上，匹配不到是 NaN
right join：保留右表全部行
inner join：只保留两边 key 都有的交集
outer join：保留并集（两边所有 key），缺的补 NaN

记住：join 的选择会改变样本数量与缺失结构（对后续建模很关键）。

Source: Combining datasets, from Duke MIDS Practical Data Science course IDS 720 by Kyle Bradbury and Nick Eubank.

Feature Selection

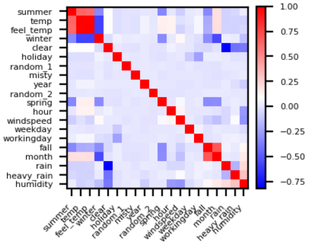
What is Feature Selection?

- The process of selecting the most relevant features for the analysis.
- Reduces dimensionality, improves interpretability, and enhances model performance.
  降低维度，提高可解释性，并增强模型性能。

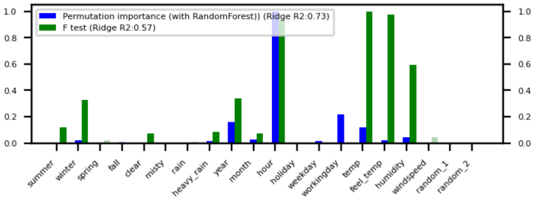
Common Methods:

- **Correlation:** Identifies features highly correlated with the target variable.
- **Variance Threshold:** Removes features with low variance.低方差
- **Feature Importance:** Uses algorithms like Random Forest to rank feature importance.

A heatmap showing feature correlations



A bar chart ranking features by importance



Dimensionality Reduction

What is Dimensionality Reduction?

- Reducing the number of features while preserving important information.

Why is it Important?

- Reduces computational complexity.
- Improves model performance by removing noise.
- Visualizes high-dimensional data.

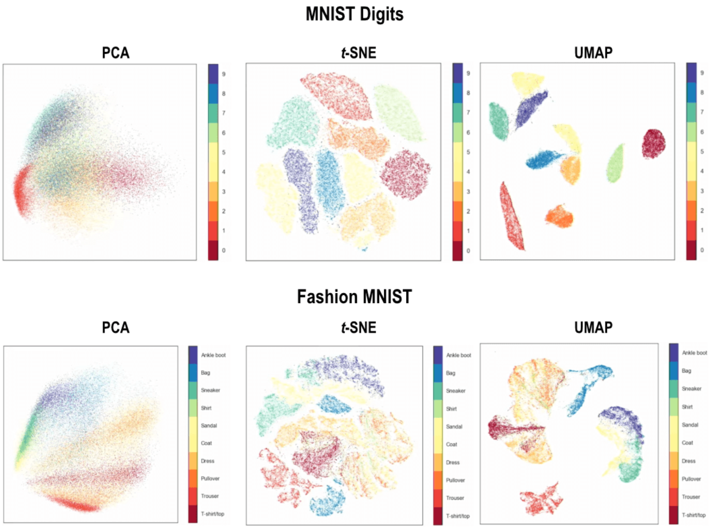
Common Techniques:

- PCA (Principal Component Analysis).
- t-SNE (t-Distributed Stochastic Neighbor Embedding).
- UMAP (Uniform Manifold Approximation and Projection).

Methods for Dimensionality Reduction

- **Principal Component Analysis (PCA):**
  - Projects data onto a lower-dimensional space.
  - Retains maximum variance.
- **t-SNE:**
  - Non-linear dimensionality reduction technique.
  - Focuses on preserving local relationships in data.
- **UMAP:**
  - Preserves local and global data structures.
  - Suitable for clustering and visualization.
    聚类

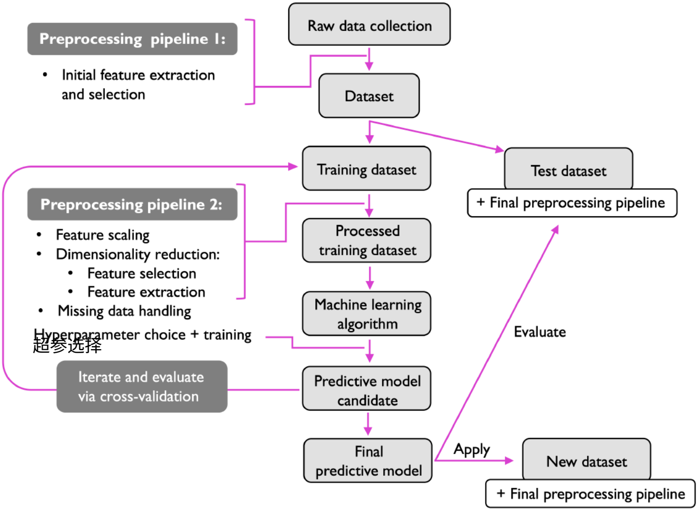
Dimensionality Reduction



Source: https://meta.caspershire.net/umap/

The Complete ML Workflow

The Complete ML Workflow



Source: STAT 451 – Introduction to Machine Learning and Statistical Pattern Classification by Sebastian Raschka